

respatial

Prior art and related tools: what exists, and how this differs

September 20, 2026

1 Scope and method

A survey of free and commercial tools that overlap with the individual features of **respatial** (per-activity re-panning, per-bin stage expand/contract, rigid scene rotation, per-bin spatial measurement, before/after verification, inverse-distance gain). Method: web search plus prior knowledge, September 2026; product claims are as described on the vendors' pages or in roundups found in search. The plugin space is large, so this is not exhaustive (Section [5](#)).

2 Overview

Feature	Closest existing tools	Key difference
Rotation	LPanner (free, VST3, $\pm 50^\circ$); Noisebud 360 (free, phase rotation); Rawton “Rotation” (M/S quadrature); ambisonics scene rotation (e.g. ambdec toolchains)	Ours is a rigid <i>position-space</i> ILD remap (energy-normalized, phase untouched, edge clipping); theirs rotate the M/S vector or phase quadrature (HAAS-like)
Stage expand/contract	Stereo wideners: Waves (Stereo-maker, S1), iZotope Ozone Imager, Mensla Stereo Tool, Voxengo Spatifier, Nugen Stereoizer, Polyverse Wider (free)	Wideners scale the side signal (global width %); ours solves γ so $p95(d) = S/2$ per bin, with energy normalization and blend
Analyze (activities $\rightarrow$ re-pan)	Demucs (free, Meta); Spleeter (free); MISST (free GUI)	Theirs separate <i>fixed</i> source types (vocals/drums/bass/other) and do not re-position; ours detects arbitrary timbral activities (NMF, shared D), fingerprints each spatially, and re-pans it
Measure / verify	CMUSE ILD calculator (forward model: angle $\rightarrow$ ILD); Ozone Imager / Mensla vectorscopes (visual, time-domain)	Nothing found that prints a <i>per-bin position distribution</i> of a recording (median/p5/p95/range, histogram, bias), suggests <code>--rotate/--stage</code> values, or checks a transformation before/after numerically
Inverse-distance gain	Praat “Advanced Stereo Panner” (distance modeling for placing a mono source); CMUSE head-diffraction ILD model	Those place a <i>known</i> source; ours applies the stage-model gain to <i>relocated</i> content in an existing mix

Adjacent, different goal: *upmixing* to more channels (UpM1xer, Corr Audio — commercial; coderynx/spatial-audio-upmixer — free Python).

3 Feature-by-feature notes

3.1 Rotation

“Rotating the stereo field” is a real plugin category. LPanner (<https://github.com/liquid1224/LPanner>, MIT, VST3, Windows) offers “Rotation: -50° to $+50^\circ$ ” plus width 0–200%; Noisebud 360 (free) rotates “the entire stereo signal, or just the left or right channel” as a *phase* rotation (“similar to the HAAS technique but without any time delay”); Rawton Forge’s “Rotation” is a stereo *quadrature* processor. In ambisonics, rotating the encoded scene is a standard operation (rotation matrices on the ambisonic coefficients) — but that domain has an explicit scene representation, which plain stereo lacks.

The mechanism difference matters: M/S-vector and quadrature rotation mix phase and amplitude in the mid/side plane (a different, timbre-altering operation), whereas `--rotate` remaps each bin’s ILD through the equal-power position model ($d' = \text{clip}(d + \theta)$), preserving energy per bin and phase exactly. None of the found tools pair the rotation with a measurement of how much the recording is biased.

3.2 Stage expand/contract

Stereo wideners/imagers are a crowded, mature category (free: Ozone Imager, Polyverse Wider; commercial: Waves, Mensla, Voxengo, Nugen, BOOM Library STEREO LAB, Red Rock Sound FUSE, DDMF StereoerS). Their width control scales the side signal; the result is a global width percentage with no notion of *where* the content sits. `--stage` instead targets a position-width (p_{95} of the deviation distribution) and is expressed in degrees of stage, which composes with rotation.

3.3 Analyze (decomposition to re-panning)

Demucs (<https://github.com/facebookresearch/demucs>, free, Meta) and Spleeter (free) are state-of-the-art *music source separation* into fixed source types; MISST is a free local GUI around Demucs. They never inspect the L/R spatial distribution and do not re-position stems. NMF-based audio source separation is a well-studied research area (Lee & Seung; see the design notes), but no widely used free tool was found that detects *arbitrary* timbral activities in a stereo mix, fingerprints each spatially (median ILD / circular-mean ICPD), and re-pans them. Praat’s “Advanced Stereo Panner” script (free) places a mono source using ILD, ITD, spectral cues and distance modeling — closest to our rendering + `--depth` model, but for placing one source, not repositioning a mix.

3.4 Measure and verify

Found: the CMUSE ILD calculator (web; estimates ILD from angle, frequency, distance and head size — the *forward* direction of our $d \leftrightarrow$ ILD map); an interactive ITD/ILD positioning demo (Maximilian Klein); and visual stereo analyzers (goniometers/vectorscopes) inside Ozone Imager and Mensla Stereo Tool. Nothing found that (a) reports a per-bin *position distribution* of a recording (median/mean/ p_5/p_{95} / range, ILD contrast, 10° histogram), (b) converts that into concrete `--rotate`/`--stage` suggestions, or (c) performs a numerical before/after check of a transformation (per-bin shift on the common active set) — the `measure` and `verify`/`--check` functionality.

3.5 Inverse-distance gain

The stage-model gain $g = r(d_0)/r(d_1)$ (listener fixed, source slides) is our own application of the inverse-distance law to relocated content. Distance modeling exists in source-placement tools (Praat panner) and in head-related ILD calculators (CMUSE), but not, as far as found, in stereo enhancement of existing recordings.

4 Bottom line

The individual concepts exist: “rotate the stereo field” and “stereo width” are real plugin categories (free and commercial), and AI stem separation is a mature free space. The *combination* does not appear in any found tool: measuring a recording’s per-bin spatial distribution numerically,

translating it into concrete rotate/stage parameters, applying a position-space rigid shift or target-width ILD warp with energy and phase preservation, and verifying the result against the command's intent.

5 Caveats

- Plugin space is large; this survey is web-search + memory and not exhaustive. A niche tool may exist.
- Vendor descriptions were not all independently tested; mechanism claims (e.g. LPanner's rotation math) are from their documentation.
- "Does not exist" claims are as-of the survey date.

6 References

1. LPanner (stereo imaging and rotation VST3, MIT): <https://github.com/liquid1224/LPanner>
2. Noisebud 360 (free phase-rotation plugin), review: <https://audiolabry.com/noisebud-360-creative-phase-rotation-plugin/>
3. Rawton Forge, "Rotation" (stereo quadrature processor): <https://isotonikstudios.com/product/rotation-by-rawton-forge/>
4. Mensla Stereo Tool (M/S shaper + vectorscope): <https://mensla.com/plugins/stereotool>
5. Waves stereo imaging plugins: <https://www.waves.com/plugins/stereo-imaging>
6. D. D. Lee and H. Seung, "Algorithms for non-negative matrix factorization," NIPS 2000: https://proceedings.neurips.cc/paper_files/paper/2000
7. Demucs (Meta, music source separation): <https://github.com/facebookresearch/demucs>
8. MISST (local GUI for Demucs): <https://github.com/Frikallo/MISST>
9. spatial-audio-upmixer (free Python, stereo → surround/Atmos): <https://github.com/coderynx/spatial-audio-upmixer>
10. UpM1xer (stereo → Dolby Atmos upmixer): <https://max1musaudio.com/plugins/upmixer>
11. Corr Audio (immersive upmix/spatialization plugins): <https://corraudio.com>
12. Schoeps Mono Upmix Plugin (mono → stereo/LCR): <https://schoeps.de/en/products/accessories/plugins/mono-upmix-plugin.html>
13. Voxengo Spatifier (mono → stereo): <https://www.voxengo.com/product/spatifier/>
14. CMUSE ILD calculator (angle/frequency → ILD): <https://www.cmuse.org/ild-stereo-binaural-calculator/>
15. "Advanced Stereo Panner" (Praat script; ILD/ITD/distance placement): https://mashav.com/sha/praat/scripts/Advanced_Stereo_Panner.html

16. M. Klein, “Cross-Talking Stereo ITD/ILD Tester” (interactive demo): <https://maximilianklein.github.io/showcase/spatial-audio/>
17. BOOM Library STEREO LAB (mono $\rightarrow$ stereo, L/R panning, ITD): <https://www.boomlibrary.com/sound-effects/stereolab/>
18. DDMF Stereoerets (stereo field plugin): <https://ddmf.eu/stereoerets-stereo-field-plugin/>
19. Red Rock Sound FUSE Stereo Image (M/S): <https://redrocksound.pro/en/stim-pc/>